Part A: Probability & Conceptual Foundation (Theory)

1. What is Conditional Probability?

Conditional Probability is the probability of an event occurring given that another event has already occurred. It helps us update our prediction based on available information.

Formula:

P(A|B)=P(A∩B)P(B)P(A|B)=\frac{P(A \cap B)}{P(B)}P(A|B)=P(B)P(A∩B)

where:

- P(A|B)P(A|B)P(A|B) = Probability of A given B
- P(A∩B)P(A \cap B)P(A∩B) = Probability of both A and B occurring
- P(B)P(B)P(B) = Probability of B occurring

In spam detection, conditional probability can be used to determine the probability that a message is spam given certain message features.

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2. Explain Bayes' Theorem and its Importance in Classification Problems

Bayes' Theorem is a mathematical formula used to calculate the probability of an event based on prior knowledge of related conditions.

Formula:

P(A|B)=P(B|A)×P(A)P(B)P(A|B)=\frac{P(B|A)\times P(A)}{P(B)}P(A|B)=P(B)P(B|A)×P(A)

where:

- P(A|B)P(A|B)P(A|B) = Posterior Probability
- P(B|A)P(B|A)P(B|A) = Likelihood
- P(A)P(A)P(A) = Prior Probability
- P(B)P(B)P(B) = Evidence

Importance in Classification:

- It helps predict the class label based on observed features.
- It forms the foundation of the Naive Bayes Classifier.
- It is efficient for spam detection, text classification, and email filtering.

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3. What Assumptions does the Naive Bayes Classifier Make?

The Naive Bayes Classifier assumes that all input features are conditionally independent of each other given the target class.

For example, in spam detection:

- Message length
- Number of URLs
- Number of special characters

are assumed to contribute independently to whether a message is spam or legitimate.

Key Assumptions:

1. Feature Independence.
2. Equal contribution of features.
3. Probabilities are estimated from training data.

Although this assumption is often unrealistic, Naive Bayes performs well in many real-world classification tasks.

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4. Explain the Working Principle of:

A) K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a distance-based supervised learning algorithm. It classifies a new data point based on the majority class among its nearest neighbors.

Working:

1. Choose the value of K.
2. Calculate the distance between the new data point and all training samples.
3. Select the K nearest neighbors.
4. Assign the class that appears most frequently among those neighbors.

Advantages:

- Simple and easy to understand.
- No training phase required.

Disadvantages:

- Computationally expensive for large datasets.
- Sensitive to feature scaling.

B) Support Vector Machine (SVM)

Support Vector Machine (SVM) is a margin-based classification algorithm that finds the optimal boundary (hyperplane) separating different classes.

Working:

1. Identify support vectors near the decision boundary.
2. Construct a hyperplane that maximizes the margin between classes.
3. Classify new observations based on which side of the boundary they fall.

Advantages:

- Effective for high-dimensional data.
- Works well with clear class separation.

Disadvantages:

- Requires feature scaling.
- Can be computationally expensive for very large datasets.

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5. Compare Distance-Based, Probabilistic, and Margin-Based Classifiers

Aspect	**KNN (Distance-Based)**	**Naive Bayes (Probabilistic)**	**SVM (Margin-Based)**
Principle	Uses distance between data points	Uses probability calculations	Uses maximum margin separation
Training Speed	Fast	Very Fast	Moderate
Prediction Speed	Slow	Fast	Fast
Interpretability	Medium	High	Medium

Aspect	KNN (Distance-Based)	Naive Bayes (Probabilistic)	SVM (Margin-Based)
Works with Large Data	Moderate	Excellent	Good
Handles Non-linear Data	Yes	Limited	Yes (with kernels)
Best Use Case	Similarity-based classification	Text and spam classification	Complex classification problems

Conclusion

- **KNN** classifies data based on similarity and distance.
- **Naive Bayes** uses probability theory and Bayes' Theorem for prediction.
- **SVM** creates an optimal decision boundary by maximizing the margin between classes.

For spam message classification, Naive Bayes is commonly used due to its probabilistic nature, while SVM often provides high accuracy and KNN serves as a useful baseline model.